

The Acoustic Plenum and the Photon

Lien (2026) in the light of Docs. 267, 290, 340, A265 and R128

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The Acoustic Plenum and the Photon

Abstract

C. C. Lien (Zenodo, DOI 10.5281/zenodo.21800087, 8 September 2026) interprets space as a compressible superfluid, derives the electron mass from the temperature of the cosmic microwave background, adopts a mass formula for the muon and tau, equates the neutrino mass with $k_B T_{\text{CMB}}$ and describes the photon as a longitudinal shockwave with a transverse wake. This document examines the paper as a case study with a verification script (20/20). The numerical values are reproducible **[K]**. None of the load-bearing claims survives the check: the electron hit lies 88 standard deviations from the measured value and is not significant in the null test, the lepton formula is known, the neutrino mass is incompatible with oscillation data, the redshift would be frequency-dependent, and the photon would have three instead of two polarisations **[X]**. The cross-check shows how FFGFT proceeds at the same points, without repeating those contents.

Status markers

- [B]** Algebraically proved.
- [K]** Numerically confirmed (verification script).
- [S]** Structurally plausible, open.
- [X]** Examined and rejected.
- [Q]** Quoted verbatim from the source (Lien 2026).

1 The source

Lien replaces empty space with a medium:

[Q]"By treating the spatial manifold as a continuous compressible superfluid, we derive the foundational particle phase-states (electron, muon, tau, and neutrino) from the Cosmic Microwave Background (CMB) acoustic baseline." (Conclusion)

The background is stationary, and redshift is not expansion:

[Q]"Cosmological redshift is therefore not temporal expansion; it is Acoustic Pulse Broadening – the observation that waves of higher energetic frequencies decay faster than those of a lower energetic frequency when traversing the same medium." (Section II.A)

The photon is not a particle:

[Q]"Photonic emission is therefore not a discrete point-particle phenomenon. It is a primary longitudinal acoustic shockwave [...] which structurally interacts with the azimuthal shear of its source to continuously generate an orthogonal transverse spiral wake." (Section V)

The central formulas are

$$f_e = 2\pi(\alpha^{-1})^4 \frac{k_B T_{\text{CMB}}}{h}, \quad (1)$$

$$m_n = m_e \left(1 + \frac{3}{2} \alpha^{-1} \sum_{k=0}^n k^4 \right), \quad (2)$$

$$m_\nu = k_B T_{\text{CMB}} / c^2. \quad (3)$$

2 Recalculation

All six numerical values are reproducible **[K]**: $f_{\text{CMB}} = 5.679 \times 10^{10}$ Hz, $f_e = 1.2583 \times 10^{20}$ Hz against the Compton frequency 1.2356×10^{20} Hz (+1.84 %), $m_\mu/m_e = 206.55$ (−0.10 %), $m_\tau/m_e = 3495.4$ (+0.52 %) and $m_\nu = 0.235$ meV. There are no arithmetic errors. The only question is what the numbers mean.

3 Significance

The electron hit is a miss. The ratio $m_e c^2 / (k_B T_{\text{CMB}}) = 2.176 \times 10^9$ lies 1.84 % below $2\pi\alpha^{-4} = 2.216 \times 10^9$. T_{CMB} is known to 0.02 %; the deviation therefore amounts to about 88 standard deviations **[K]**. Lien interprets it as "dynamic acoustic drag" but does not calculate this drag. A friction named after the fact is not a correction but a second free parameter.

Null test. The formula family $a\pi^b\alpha^{-k}$ with thirteen simple fractions $a, b \in \{-3, \dots, 3\}$ and $k \in \{1, \dots, 6\}$ comprises 546 formulas. Against random target values between 10^9 and 10^{10} , a hit within 1.84 % succeeds in 61 % of cases **[K]**. The hit is therefore not significant. FFGFT applies the same standard to its own grid findings (Doc. 342, 343 §G'): what chance delivers equally well is not counted.

The lepton formula is known. Eq. (2) is the formula of A. O. Barut (Phys. Rev. Lett. 42, 1251, 1979) without change **[B]**. The accuracy for muon and tau is Barut's result, not that of the plenum. For $n = 3$ the same formula would give a charged lepton at 10.3 GeV **[K]**, which is experimentally excluded; Lien's justification via Chirikov overlap remains qualitative. For the number of generations in FFGFT see Doc. 317 and 316.

4 Redshift

Damping changes the amplitude, not the frequency. Stokes damping $\alpha \propto f^2$ reduces the amplitude of a wave; its frequency is preserved. Numerically, a damped 50 Hz oscillation keeps its spectral maximum at 50 Hz **[K]**. No redshift arises in this way.

A frequency-dependent redshift is excluded. If z were tied to damping $\propto f^2$, the 21 cm line and Lyman- α would differ by a factor of 3×10^{12} **[K]**. As measured, z agrees between the two to about 10^{-5} . FFGFT went through this error itself: older corpus documents contained a chromatic redshift, which A265 marks as superseded.

Light-curve stretching. Lien mentions the $(1+z)$ stretching of supernova light curves only as an argument of standard cosmology, without explaining it. A static picture must contain it. In FFGFT it follows from $\tilde{T} \cdot m = 1$: wavelength and clock rate are booked together, and the stretching is exactly $(1+z)$ for all z (Doc. 312, A265, R128) **[B]**. By Doc. 267 all frequency and clock-rate signatures are degenerate between a static picture and expansion; a static background is therefore admissible, but only if it is achromatic. Lien's plenum is not **[X]**.

5 The photon

In vacuum, light has exactly two transverse polarisations and no longitudinal component. A massless $U(1)$ gauge field has four components, two of which are removed by gauge freedom and the constraint **[B]**. Lien's picture of a longitudinal primary wave plus a transverse wake would have three degrees of freedom **[X]**. The baroclinic torque $\nabla\rho \times \nabla P$ generates a vorticity component in the z direction, but no wave with two polarisation states. Photon counting (photoelectric effect, antibunching) is not addressed in the paper.

The argument that a particle picture would have to break up into a "quantised spray" in the far field confuses the wavefront of a classical field with the counting statistics of its quanta. In FFGFT the photon is exactly massless, the pure phase and connection anchor, and $E_{\text{bit}} = \hbar c/L$ is a photon energy (Doc. 290). No medium is needed for this.

6 Neutrino

A single mass of 0.235 meV cannot produce the atmospheric mass splitting $\Delta m_{\text{atm}}^2 \approx 2.5 \times 10^{-3} \text{ eV}^2$. For that, at least one state must weigh about 50 meV **[K]**. Equating the neutrino

mass with $k_B T_{\text{CMB}}$ is therefore excluded **[X]**. The FFGFT hierarchy of Doc. 340 satisfies this condition with its heaviest state at 49.96 meV **[K]**.

7 The shared anchor

It is striking that Lien touches the same number that is the SI anchor of the cosmic sector in FFGFT: $m_e c^2 / (k_B T_{\text{CMB}}) = 2.176 \times 10^9$ (A265). FFGFT carries this ratio explicitly as an anchor, not as a derivation; the exponent 41/4 of the bridge formula is open (P20), and the cosmic sector is deliberately bracketed (P39). Lien's candidate $2\pi\alpha^{-4}$ misses the number by 1.84 % and cannot be distinguished from chance in the null test. It therefore does not close the gap; it shows how easily approximations can be found at this point **[S]**.

8 What agrees and what does not

The paper shares two starting points with FFGFT: a stationary instead of an expanding background, and the rejection of point-like singularities. In FFGFT the smallest length $L_0 = \xi \ell_P$ secures finiteness (Doc. 180, R90); in Lien's work a density maximum ρ_E does this. That maximum, however, is circular, because the classical electron radius $r_e = \alpha \hbar / (m_e c)$ already contains the electron mass **[K]**.

The difference lies in the method. Lien introduces a medium with material constants (damping, viscosity, drag) and explains deviations after the fact through these constants. FFGFT posits a compact geometry with one parameter ξ , derives numbers and checks them against fixed criteria. Where a deviation remains, it is carried as an open bridge, not explained away.

9 Summary

Claim (Lien)	Finding	FFGFT	Status
$f_e = 2\pi\alpha^{-4} f_{\text{CMB}}$	+1.84 %, 88 σ , null test 61 %	A265, P20	[X]
Lepton formula Eq. (2)	Barut 1979, $n = 3$ excluded	316, 317	[X]
$m_\nu = k_B T_{\text{CMB}}$	incompatible with Δm_{atm}^2	340	[X]
Redshift by damping	frequency-dependent, leaves f unchanged	267, 312, R128	[X]
Photon longitudinal + transverse	three instead of two degrees of freedom	290	[X]
Stationary background	admissible if achromatic	267, R128	[S]

The paper combines known formulas, a friction named after the fact and an intuitive medium into a picture that appears closed. The recalculation shows that the numbers are correct, but that the interpretation fails against measured data at every load-bearing point. The usable core, a static background without singularities, is already realised differently and achromatically in FFGFT.

Verification script. 2/python/Dok379_Skripte/pruef_379_lien_photon.py (20/20): recalculation (A), significance and null test (B), measured data (C), FFGFT cross-check (D). Two inputs are literature values and marked as such: the agreement of z between 21 cm and Lyman- α , and the DES value $b = 1.003 \pm 0.011$.

Source document. Lien, C. C.: *Electromagnetic Illusion: Photonic Emission as Orthogonal Acoustic Shockwaves in a Compressible Superfluid Plenum*. Zenodo (2026), DOI 10.5281/zenodo.21800087. Barut, A. O.: *Lepton Mass Formula*. Phys. Rev. Lett. **42**, 1251 (1979).